

Toward an Independent AthenaK Reproduction of CGL–Landau-Fluid Self-Organization in High- β Turbulence

May 30, 2026; WO1 method update September 29, 2026

Abstract

Pressure anisotropy may reorganize collisionless, high- β turbulence before microscale instabilities provide an effective collisional closure, but the numerical evidence for this “magneto-immutable” response is demanding to reproduce. We define an independent AthenaK reproduction of the CGL–Landau-fluid (CGL–LF) simulations of Majeski, Kunz, and Squire (2024, MKS24). The WO1 update adds resolved operator and wave checks, finite-rate limiter verification, block/rank comparisons, and multilevel conservation tests. The reproduction targets sixteen unique source-mapped CGL–LF calculations spanning active/passive pressure feedback, forcing geometry, beta, heat-flux scale, limiter scattering, and scale separation. Several dimensionless published products admit checksum-qualified digitized comparisons, whereas absolute spectral ordinates remain blocked pending a documented transform or original numeric data; the hybrid-kinetic comparison is outside a direct CGL–LF recalculation. The pre-replacement production lineage is retained only as archival pipeline evidence. In the later **E02-modal-driver** epoch, the replacement-driver **R16** lineage reaches exact $t = 10$, and its production-window analyzer passes. A subsequent source-to-code audit found that E02 forcing semantics are not exact; preserve E02 as pipeline and cost evidence only. The figures retained here predate WO1; corrected production comparisons are not established, and passive CGL roles are disabled pending a thermal-energy redesign.

1 Introduction

In a weakly collisional plasma, conservation of the adiabatic invariants can make changes in magnetic-field strength energetically expensive: field amplification and reduction generate a pressure anisotropy that acts back on the turbulent strain. A central question is whether this feedback suppresses the field-strength-changing motions before mirror and firehose regulation dominates the dynamics. MKS24 addressed this question with driven high- β CGL–LF turbulence and found strong active/passive contrasts in instability exposure, transfer, and scale-dependent organization.

The numerical claim is consequential because it combines non-local heat-flux closure, threshold-activated anomalous scattering, and turbulent statistics in an elongated mean-field box. An independent reproduction must therefore match the physical experiment before interpreting a new parameter scan. In particular, MKS24 varies the anomalous limiter rate after a microinstability threshold is crossed; it does not scan a uniform background collision frequency. We retain the latter experiment as later work, rather than confusing it with the published target.

This manuscript retains the illustrated report originally maintained on branch **CGL–STS–LF**, with a dated WO1 method and acceptance update. It documents the AthenaK CGL–LF method, preserves method-level validation figures, states the MKS24 comparison contract, and records the production and reference-data boundaries needed for defensible claims.

The current method results are the WO1 acceptance tests listed below, including resolved linear-wave convergence on the Squire et al. Figure 17 background. The retained May 2026 figures are historical and do not verify the revised collision schedule, coefficient, forcing update, or tighter error bounds. The reduced nonlinear figures exercise analysis and plotting paths only: they are not standard-resolution, steady-state, or quantitative MKS24 reproduction results. The current execution and reference-admission status is maintained in `docs/cgl_lf_mks24_reproduction_implementation_plan.md` and in the Sphinx CGL-LF validation guide.

2 MKS24 Target Contract

The directly reproducible target is the CGL-LF simulation component of MKS24. We use a fully periodic domain $[L_x, L_y, L_z] = [1, 1, 2]$, with the background magnetic field along z , standard resolution $192 \times 192 \times 384$, and initial $\beta_0 \in \{1, 10, 100\}$ where required by the mapped published panels. The nominal closure scale is $|k_{\parallel}| = 4\pi/L_{\parallel} = 2\pi$, the background collision frequency is zero, and the published target uses hard-wall regulation at the mirror and parallel-firehose thresholds, $\beta\Delta = 1$ and $\beta\Delta = -2$. The current decks approximate that target with the explicit stiff finite rate described below. The forced calculations use an Ornstein-Uhlenbeck velocity source with fiducial injection $\dot{E}_K = 0.32\rho_0 v_A^2 L_{\perp}^3$, shell $k/(2\pi/L_{\parallel}) \in [1, 3]$, and power proportional to k^{-2} . Every production calculation must reach at least $t_f = 10L_{\perp}/v_A$; comparisons use declared late-time windows no shorter than $2L_{\perp}/v_A$.

The comparison has two deliberate limitations. First, MKS24 Figure 10 uses a hybrid-kinetic **Pegasus++** calculation and is external context unless that separate dataset or an equivalent kinetic rerun is obtained. Second, dimensionful spectral ordinate comparisons are withheld where the published material does not state the discrete Fourier normalization required for an AthenaK transform. We may therefore establish reproduction of specified CGL-LF products, but we do not claim complete reproduction of every published panel unless these boundaries are closed.

3 Conserved Variables

The hyperbolic CGL system evolves total energy and the conserved anisotropy variable

$$A = \rho \log \left[\frac{p_{\perp}}{p_{\parallel}} \frac{\rho^2}{|B|^3} \right].$$

The legacy array slot is still named `IMU`, with `IAN` used as an alias in CGL code. The Riemann solve remains in A , which avoids the small- $|B|$ pathologies associated with evolving magnetic moment directly.

The LF STS operator temporarily changes only the interpretation of the `IAN` slot:

$$A \rightarrow \mu_B = \frac{p_{\perp}}{|B|}.$$

The STS update advances (E, μ_B) with the LF flux divergence and refreshes $(p_{\parallel}, p_{\perp})$ from magnetic moment between stages. It restores A at sweep completion, before ordinary hyperbolic evolution, output, or restart.

In the current source tree the LF parabolic operator is owned by `src/diffusion/cgl_landau_fluid.*`; the CGL state and constraint logic are implemented in `src/eos/cgl_mhd.cpp` and `src/eos/cgl_physics.hpp`. The quantitative method tests use `src/pgen/tests/cgl_landau_fluid.cpp`, while the reduced MKS24 initialization and history diagnostics use `src/pgen/tests/cgl_lf_paper.cpp`.

4 Landau-Fluid Closure

The implemented closure uses the Snyder–Hammett–Dorland 3+1 heat fluxes,

$$q_{\parallel} = -\chi_{\parallel} \rho \nabla_{\parallel} \left(\frac{p_{\parallel}}{\rho} \right),$$

$$q_{\perp} = -\chi_{\perp} \left[\rho \nabla_{\parallel} \left(\frac{p_{\perp}}{\rho} \right) - p_{\perp} \left(1 - \frac{p_{\perp}}{p_{\parallel}} \right) \frac{\nabla_{\parallel} |B|}{|B|} \right],$$

with finite-collision BGK moment coefficients

$$\chi_{\perp} = \frac{2c_{\parallel}^2}{\sqrt{2\pi} c_{\parallel} |k_{\parallel}| + 2\nu_{\text{eff}}}, \quad \chi_{\parallel} = \frac{8c_{\parallel}^2}{\sqrt{8\pi} c_{\parallel} |k_{\parallel}| + (3\pi - 8)\nu_{\text{eff}}}.$$

The perpendicular denominator follows SHD97 eq. 49 and Sharma et al. (2006) eq. 12, deliberately using $2\nu_{\text{eff}}$ in place of the ν_{eff} printed in Squire et al. (2023) eq. 2.7. Thus $\chi_{\perp} \rightarrow c_{\parallel}^2/\nu$ in the strongly collisional limit. Here ν_{eff} adds the background frequency, one soft-limiter rate when an enabled threshold is exceeded, and one backup rate when backup limiting is enabled and a configured or fluid wall is exceeded. Soft and backup contributions add; strictness never enables backup limiting implicitly. In the collisionless limit, these reduce to $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$ and $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$. Here `lf_k_parallel` is the closure wavenumber $|k_{\parallel}|$, not a conductivity. In local mode, $c_{\parallel} = \sqrt{p_{\parallel}/\rho}$ is computed from the live face state. In background mode, only this prefactor is frozen to `lf_c_parallel0`; all gradients and thermodynamic variables remain live.

The face fluxes coupled into STS are

$$F_E^i = \hat{b}^i \left(q_{\perp} + \frac{1}{2} q_{\parallel} \right), \quad F_{\mu_B}^i = \hat{b}^i \frac{q_{\perp}}{|B|}.$$

5 Heat-Flux Cap

Following Squire et al. (2023), equations 3.1–3.2, the unlimited heat flux q_L is capped as

$$q = \frac{q_L q_{\text{max}}}{q_{\text{max}} + |q_L|}.$$

The implemented caps are

$$q_{\perp, \text{max}} = \sqrt{\frac{2}{\pi}} c_{\parallel} p_{\perp}, \quad q_{\parallel, \text{max}} = \sqrt{\frac{8}{\pi}} c_{\parallel} p_{\parallel}.$$

The cap is applied after computing the unlimited flux with the collision-aware coefficient, so collisions and limiter scattering suppress the flux before the smooth free-streaming bound is applied.

6 Threshold Policy and Retained Diagnostics

The numeric soft-threshold defaults are `firehose_threshold=2` and `mirror_threshold=1`, giving $\Delta p = -B^2$ and $B^2/2$. The legacy oblique/parallel alias maps to 1.4/2.0 and may not conflict with an explicit numeric choice. Enabled soft limiters require an explicit finite `limiter_nu_coll`. Backup limiting is explicit and requires at least one soft limiter; its default factors are 1 (firehose)

and 2 (mirror). The fluid wall $\Delta p \geq -B^2$ always applies. Backup walls default to $[-B^2, B^2]$, preserve $p_\perp + p_\parallel/2$, and are independent of which soft limiter is enabled.

Rates advance once per full timestep: first $\Delta p \leftarrow \Delta p \exp(-\nu_{\text{coll}} \Delta t)$, then, outside an enabled soft threshold Δp_* ,

$$\Delta p \leftarrow \Delta p_* + \frac{\Delta p - \Delta p_*}{1 + \nu_{\text{lim}} \Delta t},$$

followed by the applicable walls. With LF this occurs after the post sweep; walls alone follow the pre sweep and RK step. Without LF, rates and walls follow RK. A walls-only projection also follows the final post-AMR field and primitive refresh, without repeating rates. There is no soft scattering during primitive recovery or between LF stages. `limiter_hardwall=true` is rejected. The paper decks use an explicit stiff finite rate 10^{10} ; approach to the soft threshold remains asymptotic, unlike the former algebraic projection.

LF histories retain stage, floor, invalid-state, threshold-occupancy, heat-flux, and optional RK pressure-work diagnostics. Strict validation rejects floors and invalid states at every LF stage. The fluid wall and explicitly enabled backup walls are enforced at sweep entry and after the scheduled end-of-sweep projection. Intermediate LF crossings remain counted but are not fatal; no per-stage projection is added. Ordinary soft-threshold occupancy may be nonzero. The historical `lf_hwproj` field remains for compatibility and is zero in new runs.

Current fences and forcing

`passive=true` is disabled because the passive thermal-energy equation is inconsistent, pending WO2; retained isothermal signal-speed branches remain unit tested. LF split integration rejects inflow and user boundaries because they lack a magnetic-moment-aware IAN contract. Periodic, outflow, reflecting, and diode boundaries remain supported. CGL-aware AMR primitive prolongation is supported, but its optional LF pressure-work recording remains fenced.

The driver updates OU coefficients at the configured cadence and applies one kick before the RK state copy per cycle. The kick and zero-net-momentum correction each add their exact kinetic-energy change to total energy, preserving thermal energy. These corrections apply to all driven fluids, including two-fluid runs. Forcing is configured in `turb_driving`; the removed problem-level `forcing_mode` had no effect. Type 1 uses the z parallel axis and type 2 now generates its intended isotropic mode set.

7 Reproduction Campaign

The retained sixteen-case catalog includes passive roles that cannot execute until WO2. The campaign records below predate the September 29, 2026 WO1 update and do not qualify its revised executable.

The production matrix is frozen in `inputs/cgl_lf_paper/mks24_stage_i_manifest.json`. It contains sixteen unique CGL-LF calculations after two required alias decisions: the standard active, Alfvénic $\beta_0 = 10$ output also supplies the nominal heat-flux and $n_\perp = 192$ scale-separation roles, while the standard active, Alfvénic $\beta_0 = 100$ stiff-limiter output also supplies the published hard-wall-reference role. The separately defined active, Alfvénic $\beta_0 = 1$ deck is not executed in this matrix because the displayed MKS24 beta-one compressive-spectrum case is randomly forced.

Production is executed on Frontier through `scripts/frontier/cgl_lf_stage_i.py`, separately from the debug-only qualification utility. The production utility admits only mapped input files, archives input, executable, and utility checksums, uses the `batch` partition with its default production `normal` QOS, reserves usage against an explicitly reviewed epoch envelope, and permits

Cases	Physical variation	Published role
R02--R09	Active/passive; $\beta_0 = 10, 100$	State, PDF, transfer, alignment comparisons
R10--R11	Random compressive variants; $\beta_0 = 1, 100$	Compressive-spectrum comparison
R12--R13	Stronger/weaker heat-flux scales	Closure-scale sensitivity
R14--R15	$\nu_{\text{lim}} = 20, 200$ at $\beta_0 = 100$	Limiter sensitivity
R16--R17	$n_{\perp} = 96, 384$ around reused R02	Scale-separation comparison

Table 1: Unique AthenaK CGL–LF production calculations mapped to the MKS24 simulation panels. The manifest records each input file and the analysis roles that reuse an accepted output.

only one submitted segment at a time. The frozen R02–R17 matrix is executed sequentially under inspection, with R17 completed last to control peak storage.

The original pre-replacement R16 lineage remains archived as **E01-pre-modal-driver** pipeline evidence. Its legacy shared-I/O startup exposed a prolonged snapshot/checkpoint write, and its later rank-local lineage reached $t = 8.5$ cleanly. That lineage is not admitted for current reproduction because the replacement turbulent driver changes forcing evolution and retained restart state.

The fresh **E02-modal-driver** epoch starts from $t = 0$ with the qualified replacement driver. Its immutable production executable was built from revision `462b9dbd53e085dea46c2478b567781576d7d03e`, with SHA-256 digest `df87684e9d2b7af33b36c2757779d15de87b84ef051ca9c4489f2efa358f5c48`. Fresh R16 reaches exact $t = 10.0$, and its declared $8 \leq t \leq 10$ production-window analyzer passes. The standard-layout R02 lineage reaches exact $t = 2.5$ at the tracked **F-075** checkpoint. The high-resolution R17 timing lineage reaches exact $t = 0.10$. The subsequent **F-076** source-to-code audit found that E02 does not exactly implement the published forcing semantics. These records remain pipeline and cost evidence only; they do not support a direct MKS24 reproduction claim. No additional E02 preparation or submission is authorized.

The Frontier project root is shared with exploratory campaigns. In particular, Slurm records exploratory job `4743106` as complete while its top-level campaign record remains stale at **state = running**. That stale record must not be modified. Before each Stage I prepare and submit action, all user jobs and active shared-root campaign records are inspected; a stale record is acknowledged only after an isolation review.

8 Validation Workflow

Generate the method-validation figures from the repository root with

```
python3 scripts/cgl_lf_workflow.py full --output-dir /path/to/full-bundle
```

or with the compatibility command `scripts/run_cgl_lf_validation.sh`. The driver writes an ignored result bundle containing a manifest, readable summary, logs, histories, machine-readable validation CSV files, and figures. The five method figures included below are historical, pre-WO1 products from the retained bundle `build-cgl-implementation/cgl_lf_runs/20260524-final-full-v2`.

Each current test prints its measured value, reference or error norm, tolerance, and pass/fail status. The retained plotted points come from the 2026-05-24 CSV files and have not been regenerated using WO1.

Generate reduced nonlinear diagnostic products with

```
python3 scripts/cgl_lf_workflow.py paper-convergence
python3 scripts/cgl_lf_workflow.py paper-analyze \
  --output-dir /path/to/paper-convergence-bundle
```

The representative nonlinear figures below are historical products from the retained 20260525-paper-convergence-pressure-work-v1 bundle. Its cases run only through $t = 0.02$ on reduced grids and therefore test product plumbing, not statistically saturated turbulence or paper agreement.

9 Current WO1 Acceptance (September 29, 2026)

These measurements use double-precision CPU and MPI builds.

Check	Acceptance	Measured result
Short field/oblique/eigen waves	normalized Fourier error $\leq 10^{-3}$; reference change ≥ 10 tolerances	nine wave families pass
Long wave ladders	$N_x = 64, 128, 256, 1.25$ periods; order ≥ 1.8 weakly damped, ≥ 1 LF-dominated	orders 1.95–2.49; finest errors $< 5.2 \times 10^{-4}$
Parallel/perpendicular decay, $\nu = 0, 10$	$\chi k^2 T = 1$; amplitude tolerance 0.003	208–649 cycles; errors $2.08\text{--}2.71 \times 10^{-4}$
Mirror/firehose relaxation	cellwise exponential background plus finite BE and walls	relative error $\leq 10^{-12}$; isotropic pressure conserved
Two-dimensional 30° field	k along x/y ; one e-fold; amplitude tolerance 0.005	errors 0.0853%/0.0830%
Block/MPI decomposition	one block versus 2×2 ; one/four ranks	sampled physical states bit-wise identical
Two-level SMR, full STS path	amplitude tolerance 0.01; energy drift $< 2 \times 10^{-13}$	40 blocks, 70 cycles; error 0.2058%; energy drift about 7.1×10^{-15}
Gradient/cap checks	finite-volume response, flux sign and free-streaming bounds	retained dedicated operator checks

The wave/decay/limiter tests are in `tst/test_suite/cgl/test_cgl_lf_acceptance_cpu.py`. The oblique and SMR checks are in `tst/test_suite/cgl/test_cgl_lf_oblique_decay_cpu.py`, with MPI counterparts in `tst/test_suite/cgl/test_cgl_landau_fluid_mpicpu.py`. The multi-mode oblique IVP uses 1.25 Alfvén periods; individual eigenmodes use their supplied eigenfrequency. All wave checkers require a reference change large enough to reject a frozen state. Negative tests reject ineffective short references and disabled LF in compressive eigenmodes.

The decay comparison independently diagonalizes the two-temperature linear operator with background isotropization. Its acceptance window distinguishes swapped parallel/perpendicular coefficients, a factor-two perpendicular coefficient change, and the obsolete $+\nu$ denominator. The cellwise limiter oracle is exact for uniform LF states or spatially varying zero-advection pure-CGL states; nonuniform LF stress cases retain separate transport admissibility checks. Limiter tests span $\nu_{\text{lim}} \Delta t = 0, 1, 10^{10}$, background decay, and the different pure/LF wall ordering. SMR profiling confirms restriction, prolongation, and flux exchange on the full path.

10 Historical Quantitative Figures (May 24, 2026)

These retained figures document the pre-WO1 implementation and looser short-run guards. They are not graphical evidence for the current acceptance table, revised collision schedule, or updated collisional coefficient.

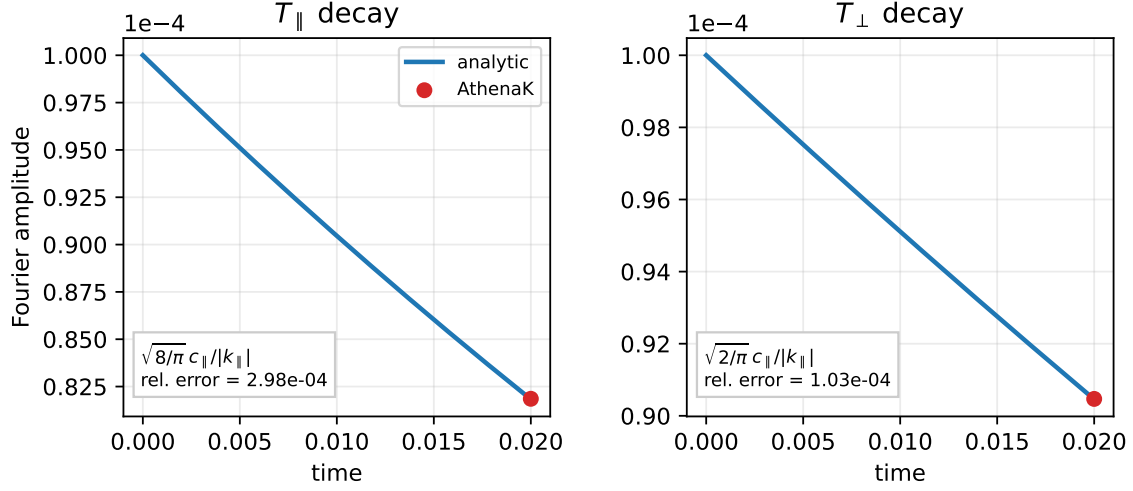


Figure 1: Historical May 24, 2026 constant-coefficient decay tests. The $T_{\parallel}$ case uses $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$; the $T_{\perp}$ case uses $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$. The red point is the final AthenaK Fourier amplitude and the blue curve is the analytic constant-coefficient decay.

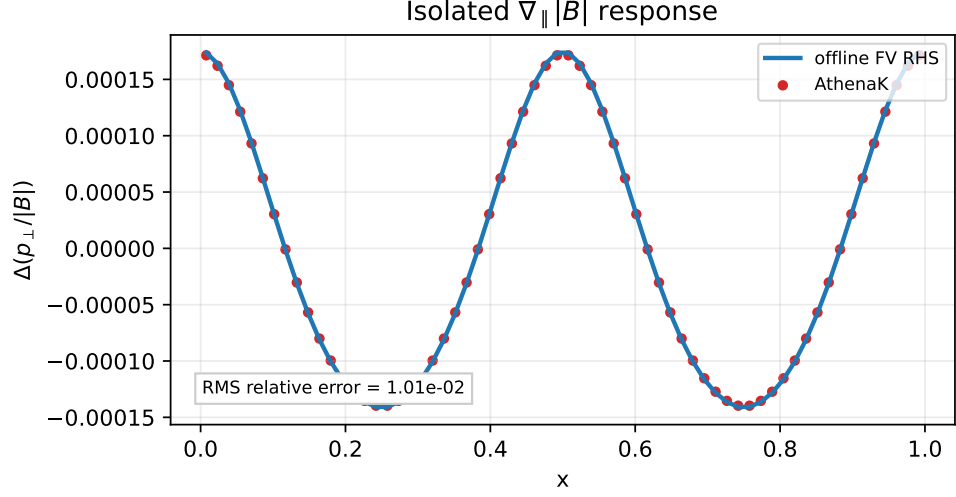


Figure 2: Historical May 24, 2026 isolated $\nabla_{||} |B|$ term in the perpendicular heat flux. The initial thermal gradients are zero; the plotted response is the short-time change in $p_{\perp}/|B|$ compared with an offline finite-volume reconstruction of the initial LF RHS.

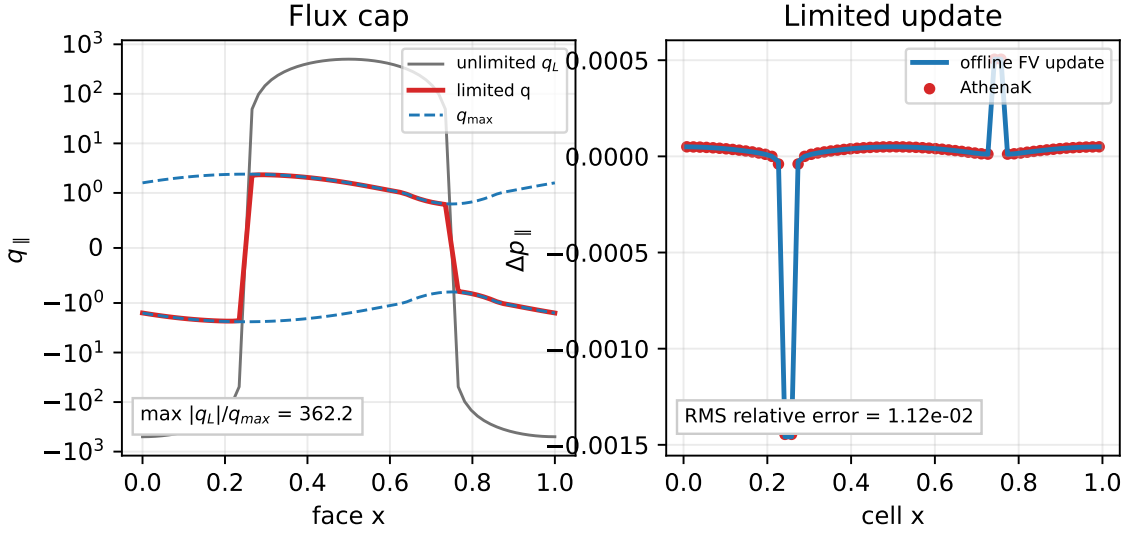


Figure 3: Historical May 24, 2026 heat-flux limiter test. The left panel shows the unlimited $q_{||,L}$, the capped $q_{||}$, and $\pm q_{||,max}$. The right panel checks that the pressure update produced by the limited flux matches the offline finite-volume update.

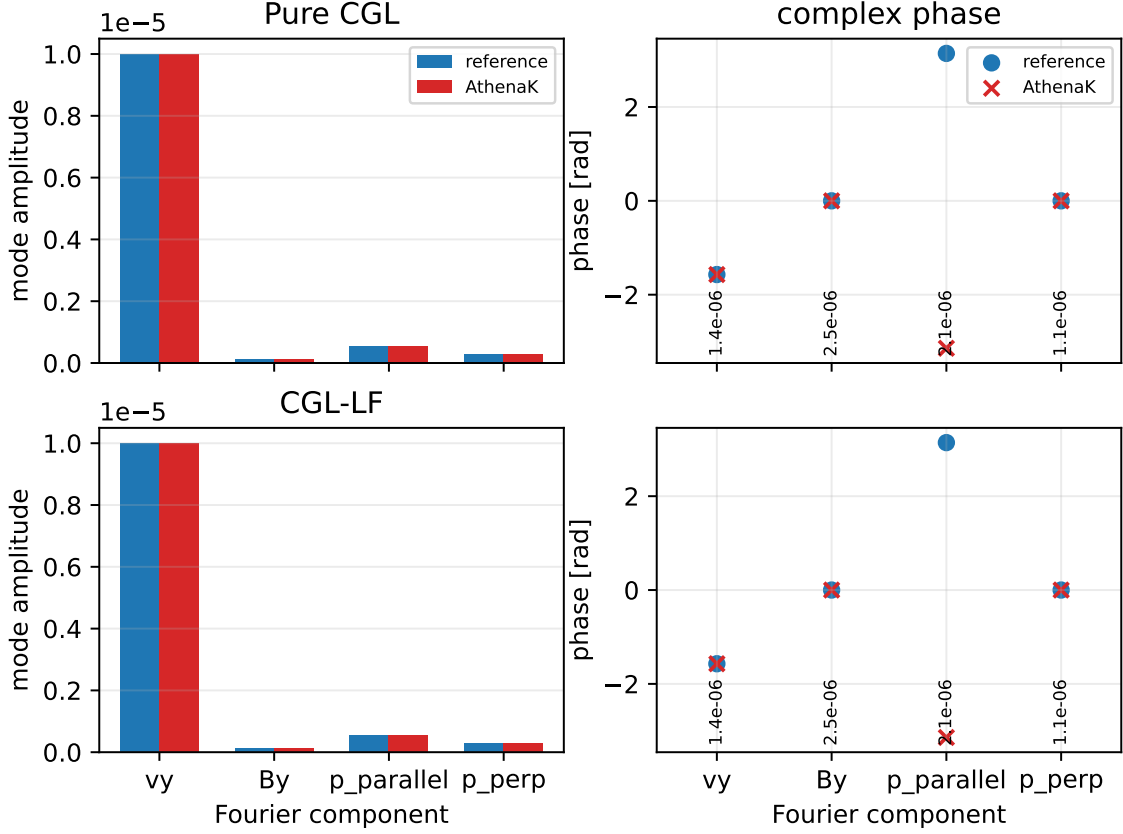


Figure 4: Historical May 24, 2026 Figure 17-style oblique linear initial-value check. The plotted quantities are the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ for the pure-CGL and CGL-LF inputs, compared with the offline linearized initial-value reference used by the quantitative pgen. The vertical phase-panel labels are per-variable relative errors.

11 Figure 17-Style Linear Initial-Value Check

The paper-style inputs use the Figure 17 background

$$\rho_0 = 1, \quad p_{\parallel 0} = p_{\perp 0} = 5, \quad u_0 = 0, \quad B_0 = (1, \sqrt{2}, 0.5), \quad k = 2\pi\hat{x}, \quad kL = |k|.$$

The test initializes a small transverse velocity perturbation, not a CGL eigenmode, and compares the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ against an offline linearized CGL–LF initial-value reference integrated with the same background state and closure coefficients. Two inputs are provided: one with the LF heat flux enabled and one with pure CGL. This checks the same ion-fluid initial-value system used for the Appendix A.2/Figure 17 comparison. The current acceptance additionally requires resolved evolution and a noninteger-period resolution ladder; the retained plot predates those checks.

12 Figure 17-Style Eigenmode Checks

The validation suite includes exact eigenmode initializations for the Alfvén, slow-like, and fast-like positive-frequency branches of the same linearized Figure 17 system. The script `scripts/generate_cgl_lf_eigenmode_inputs.py` constructs the linear CGL or CGL–LF matrix for the AthenaK variables $(\rho, u_x, u_y, u_z, B_y, B_z, p_{\parallel}, p_{\perp})$, diagonalizes it, normalizes the selected eigenvectors, and writes their complex amplitudes and eigenvalues into AthenaK input files. The pgen initializes $\text{Re}[\delta U_k \exp(ikx)]$ and compares the final Fourier mode with $\delta U_k \exp(\lambda t)$, where λ is the supplied eigenvalue. The current tests require the stated short-run tolerance and long-run convergence. They test the implemented ion-fluid equations; collisional and nonpropagating branches and electron-pressure comparisons remain outside these six eigenmode cases.

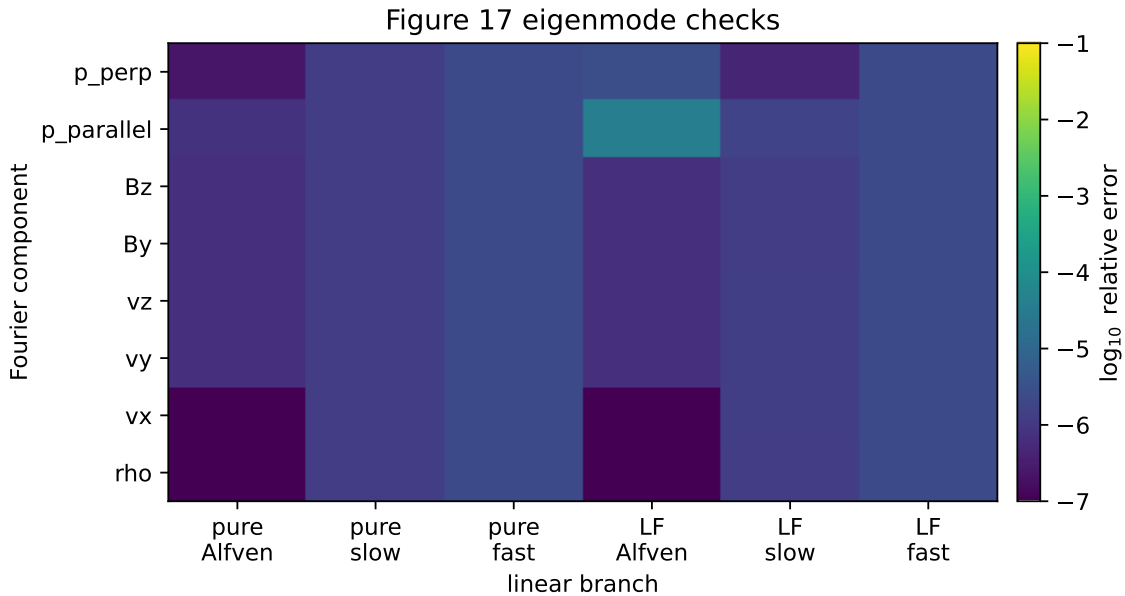


Figure 5: Historical May 24, 2026 eigenmode checks on the Figure 17 background. Each column is a pure-CGL or CGL–LF Alfvén, slow-like, or fast-like positive-frequency branch. The color scale shows the complex Fourier relative error in each primitive or magnetic component after a short AthenaK evolution compared with the supplied linear eigenmode prediction $\delta U_k \exp(\lambda t)$.

13 Reduced MKS24-Oriented Analysis Products

The `paper-convergence` workflow supplies short, reduced forced turbulence bundles used to validate nonlinear diagnostics and rendering. The figures in this section retain the pre-WO1 analysis of snapshots from `20260525-paper-convergence-pressure-work-v1`. The bundle contains three resolution-path cases and heat-flux/policy variants, all with clean LF safety diagnostics. Because these cases terminate at $t = 0.02$, their curves must not be interpreted as steady turbulent spectra or MKS24 comparisons.

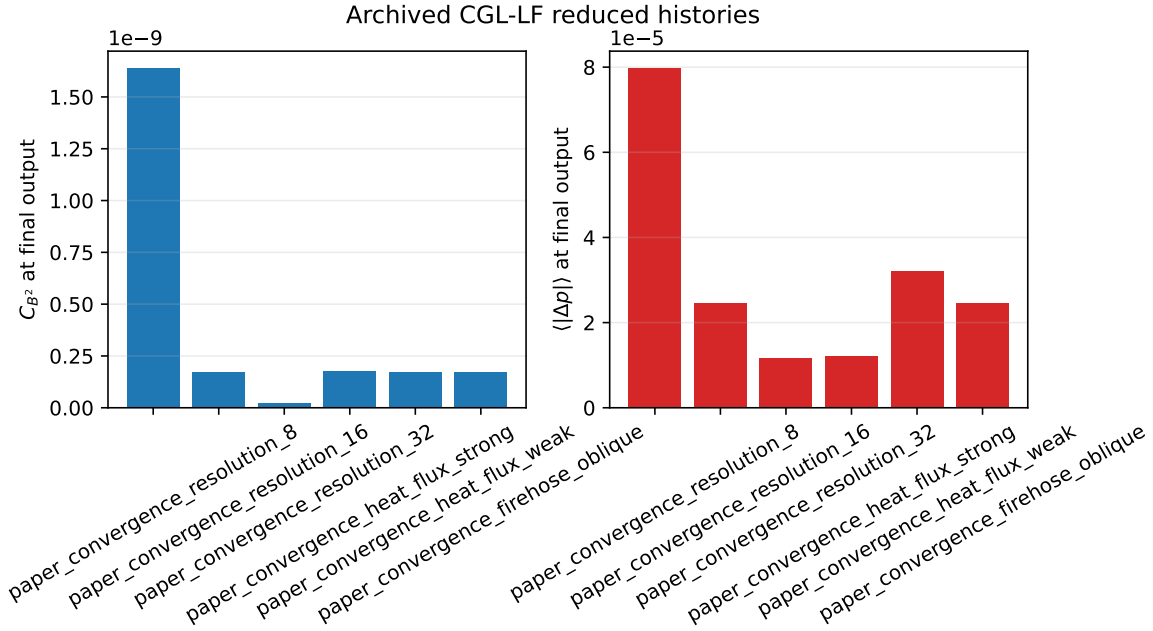


Figure 6: Historical May 25, 2026 reduced history-product rendering for the nonlinear analysis path. The plotted global summaries verify ingestion of archived history data across the reduced case matrix; they do not establish converged physical trends.

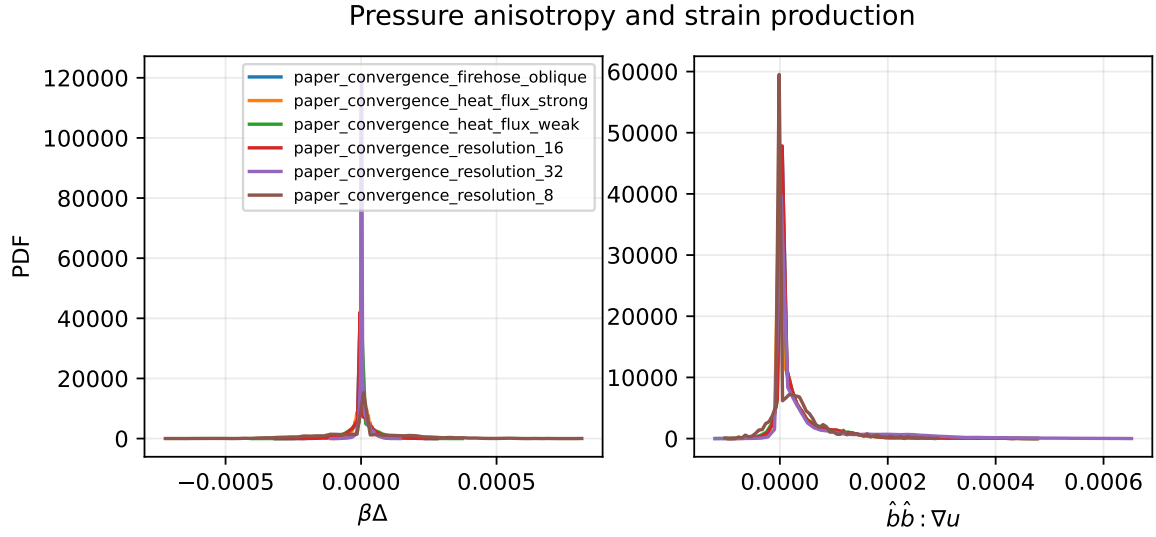


Figure 7: Historical May 25, 2026 reduced pressure-anisotropy and local-strain PDF rendering. These products expose $\beta\Delta$ and $\hat{\mathbf{b}}\hat{\mathbf{b}} : \nabla \mathbf{u}$ for later panel comparisons once authorized statistically useful runs exist.

Pressure-density joint PDF: paper_convergence_resolution_32

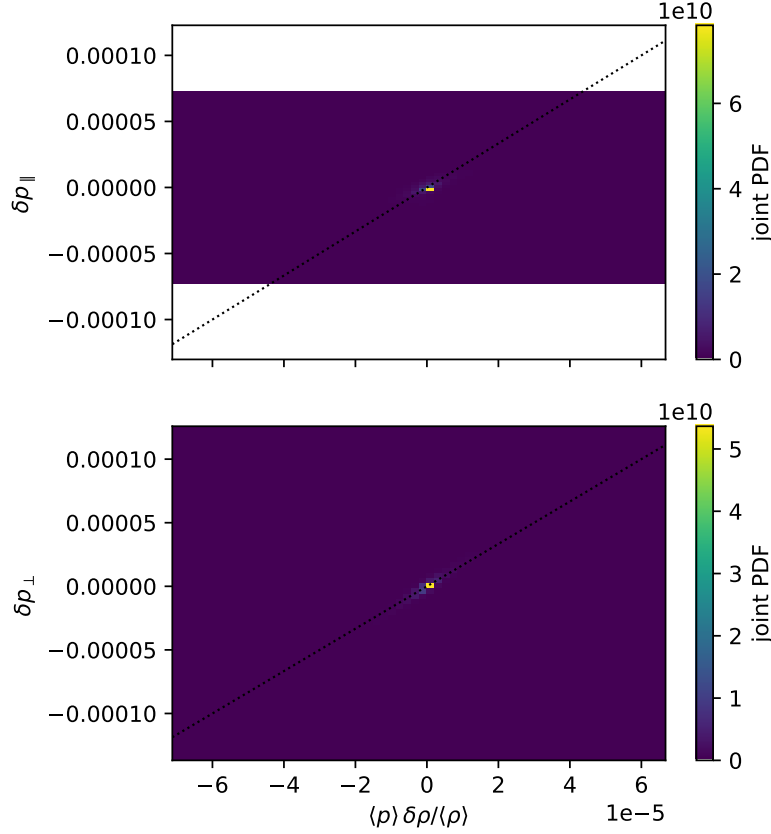


Figure 8: Historical May 25, 2026 Figure 2(a)-coordinate joint-PDF diagnostic from the $32 \times 32 \times 64$ reduced case. The analyzer now emits $\langle \rho \rangle \delta \rho / \langle \rho \rangle$ versus $\delta p_{\parallel}$ and $\delta p_{\perp}$. The admitted digitized MKS24 surface route is separate from this reduced simulation output and requires a matching standard case.

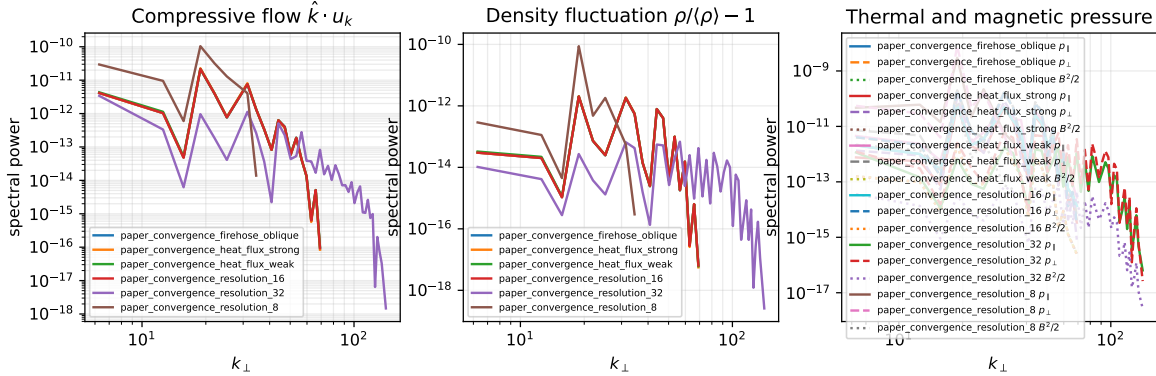


Figure 9: Historical May 25, 2026 rendering of the Figure 3/4(b)/6(a)-oriented spectral fields: compressive velocity projection, normalized density fluctuation, and thermal/magnetic-pressure products. Their plotted MKS24 absolute ordinates remain excluded from reference comparison until an external discrete Fourier-normalization mapping or original numeric data are available.

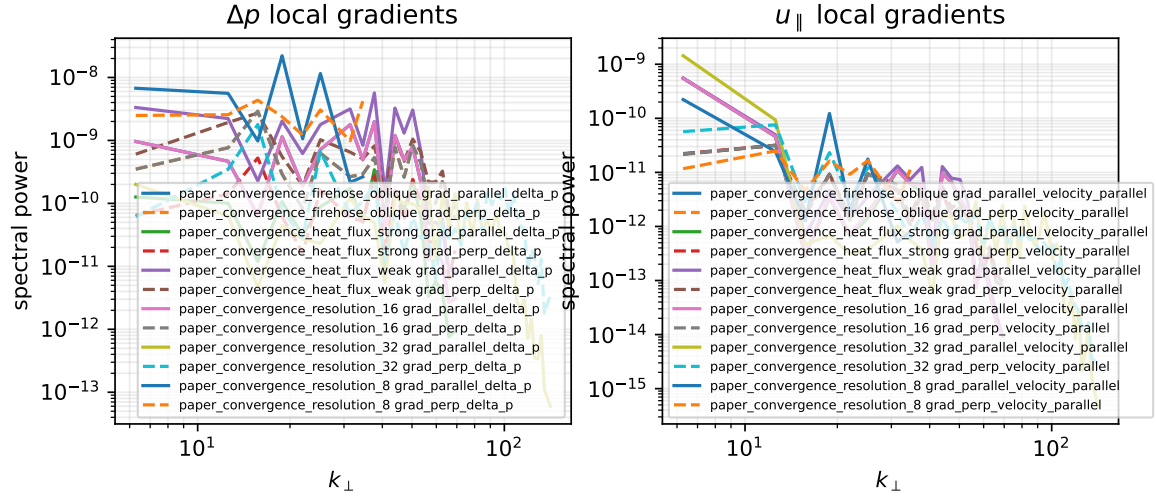


Figure 10: Historical May 25, 2026 projected pressure-anisotropy and velocity-gradient spectrum products from the reduced bundle. These are diagnostic-product checks and not admitted absolute-ordinate paper comparisons.

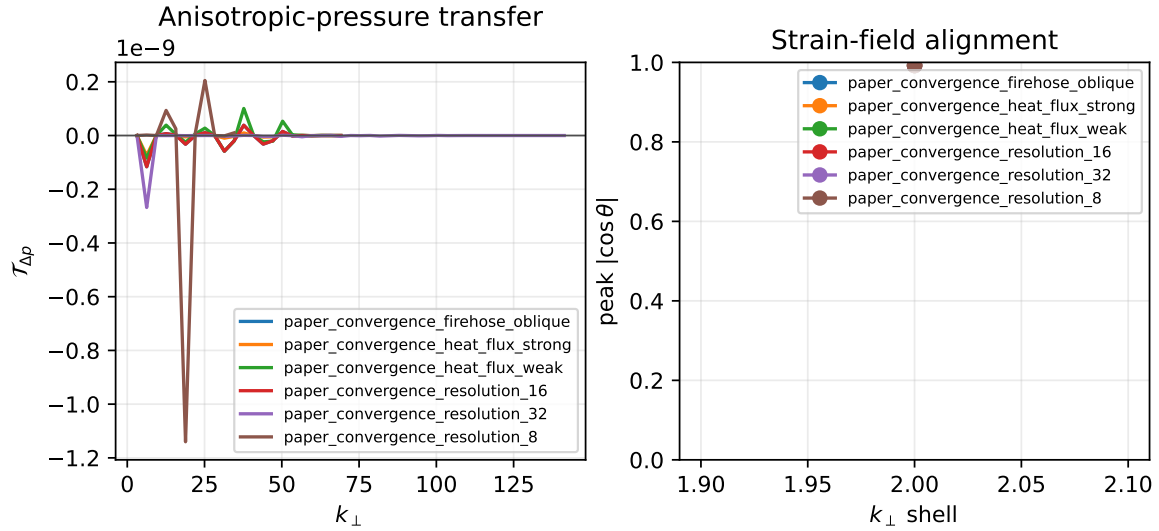


Figure 11: Historical May 25, 2026 pressure-stress transfer and alignment-product rendering. The analyzer supports the dimensionless MKS24-normalized transfer comparison and selected-shell or peak-alignment comparisons when supplied with provenance-qualified references and matching executed cases.

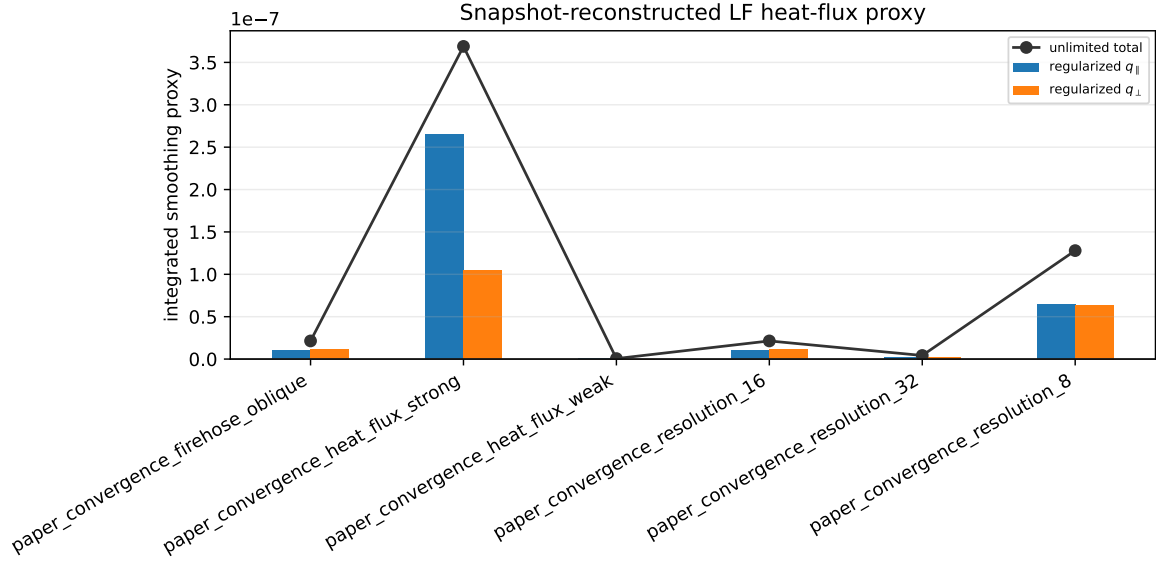


Figure 12: Historical May 25, 2026 snapshot-reconstructed heat-flux smoothing proxy for reduced cases. It is useful for interpretation of retained snapshots, but is not the applied finite-volume heat-flux contraction or a total energy closure.

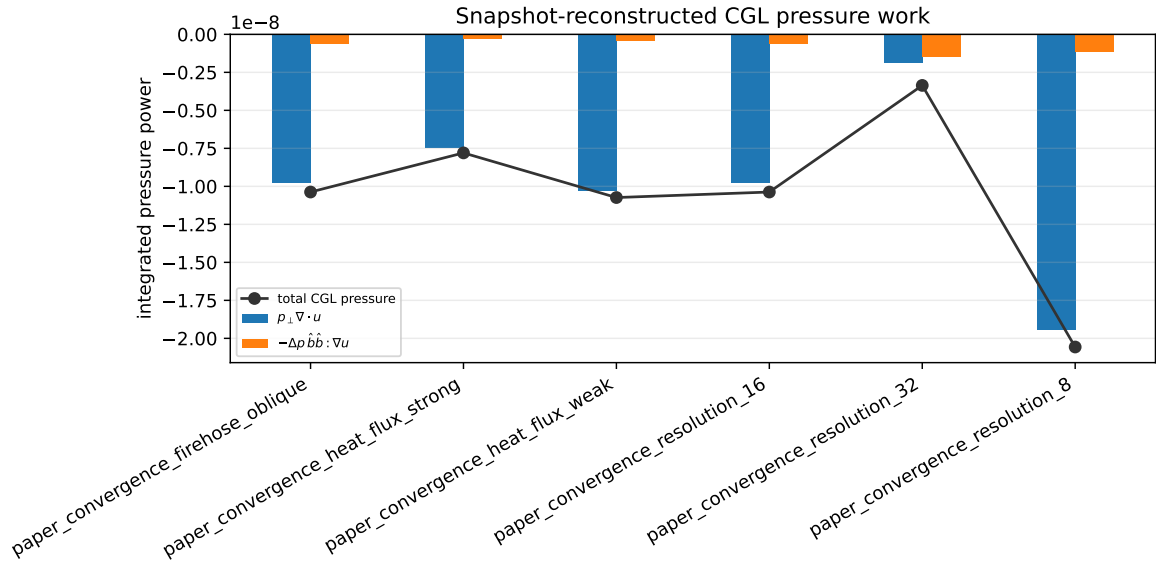


Figure 13: Historical May 25, 2026 snapshot-derived CGL pressure-work decomposition, including the anisotropic stress contribution checked against direct pressure transfer. Paper decks also retain applied pressure-traction ledger histories; neither the reduced snapshot plot nor an unevaluated standard deck establishes a production result.

14 MKS24 Reproduction Boundary

The current branch defines guarded standard, limiter-frequency, heat-flux, compressive, and scale-separation workflows and supports checksum/uncertainty-qualified external reference comparisons. The admitted published reference products presently include Figure 2(a) transformed sampled surfaces, Figure 2(b) histories, normalized Figure 4(a) PDFs, normalized Figure 5(b) eddy scales, dimensionless Figure 7 transfer ratios, Figure 8 selected-shell alignment PDFs, Figure 9 and Figure 11/12 alignment curves, and Figure 13(b),(d) dimensionless products. Because each admitted panel product retains its own provenance manifest, an accepted campaign bundle is analyzed by repeating `paper-analyze --reference-curves` for the admitted manifests; the analyzer records every manifest checksum and provenance block and rejects duplicate product identifiers across manifests. For incremental review of a single accepted case, `--allow-partial-reference-cases` evaluates only products whose cases are present in the bundle and records the omitted products; it does not authorize a missing-case panel claim. Snapshot-window selection reads the binary output time from each header before loading fields, so retained snapshots outside the declared late-time comparison window do not dominate analysis cost.

Remaining absolute spectral or strain ordinate panels are intentionally not admitted: the paper and cited donor source define shell spectra and plotted units but do not state the discrete Fourier normalization needed to map absolute digitized ordinates into AthenaK products. Those panels require author/archive numeric data, donor diagnostic code, or an explicit normalization statement. The earlier mapped production lineage is retained only as pre-replacement archival I/O-diagnostic evidence. The fresh `E02-modal-driver R16` lineage reaches exact $t = 10.0$, and its $8 \leq t \leq 10$ production-window analyzer passes. `R02` reaches exact $t = 2.5$ at the tracked `F-075` checkpoint, while the `R17` timing lineage reaches exact $t = 0.10$. The `F-076` forcing-contract audit retains these records only as pipeline and cost evidence. Quantitative MKS24 simulation-to-reference comparisons require a corrected executable, a fresh epoch from $t = 0$, matched accepted bundles, and admitted reference products; none is claimed by this manuscript version.

15 Comparison Gate Matrix

Table 2 records the evidentiary boundary for every published-panel family targeted by Stage I. An “admitted” comparison has a declared dimensionless transform and an archived digitized or analytic reference; it still requires a matched, accepted AthenaK output bundle. “Blocked” means that a necessary reference normalization or external simulation product is not available in the present reproduction archive.

16 Discussion

The reproduction is designed to distinguish a physical active-pressure response from two easier but weaker demonstrations: a correct implementation of individual LF terms and a visually plausible short turbulence transient. The current `WO1` checks establish the listed operator, linear-wave, block/rank, and SMR acceptance results; the historical figures alone do not. The guarded Stage I matrix and the comparison gates make the second requirement testable, but it cannot be resolved by the reduced diagnostic bundles shown above. Evidence for the MKS24 interpretation will require late-time active/passive contrasts in the admitted instability, transfer, alignment, and sensitivity products from accepted production data. Those passive roles are blocked pending the `WO2` thermal-energy redesign.

MKS24 panel	AthenaK cases	Quantitative gate	Status
Figure 1	R02, R06	Common-time active/passive spatial comparison; qualitative captioned interpretation only.	Awaiting runs
Figure 2(a–b)	R02--R09	Admitted sampled-surface and unstable-volume reference products.	Awaiting runs
Figure 3	R03--R05, R09--R11	Absolute compressive spectral ordinate normalization not established.	Blocked/reference
Figure 4	R02--R09	Density PDF admitted; density spectral ordinate mapping blocked.	Partial gate
Figures 5–6	R02, R04	Eddy-scale product admitted where normalized; absolute spectra blocked.	Partial gate
Figures 7–9	R02--R09	Dimensionless transfer and alignment reference products admitted.	Awaiting runs
Figure 10	none	External Pegasus++ hybrid-kinetic comparator.	External data
Figure 11	R16, R02, R17	Dimensionless scale-separation alignment product admitted.	Archived E02 only
Figure 12	R12, R02, R13, R06	Dimensionless heat-flux-scale alignment product admitted.	Awaiting runs
Figure 13	R03, R07, R14--R15	Admitted dimensionless limiter-sensitivity products only.	Awaiting runs

Table 2: Stage I panel-family comparison gates. “Awaiting runs” does not denote agreement or disagreement; it denotes absence of the required matched accepted standard-duration AthenaK bundles.

Two limitations are structural rather than statistical. The spectral ordinate panels cannot be used for absolute residual tests until the published normalization is mapped to the AthenaK transform or original numeric products are recovered. Similarly, no CGL–LF rerun is an independent reproduction of the hybrid-kinetic Figure 10 comparator. These limitations narrow the claims supported by Stage I but do not prevent quantitative evaluation of the admitted CGL–LF diagnostics.

The later weak-guide-field and background-collisionality study is therefore not part of the present results. In particular, varying uniform background collisionality would answer a different question from changing the threshold-activated limiter rate in MKS24. Treating those scans as interchangeable before the reproduction baseline is accepted would obscure both the numerical and physical comparison.

17 Conclusions

AthenaK passes the bounded WO1 acceptance checks reported here and retains nonlinear diagnostics, source-mapped MKS24 case definitions, and a checksum- and allocation-recorded Frontier production protocol. These results do not certify historical executables or complete turbulent paper reproduction. The production protocol began with the required low-resolution scale-separation case. The pre-replacement lineage is retained only as archival evidence after the turbulent-driver contract changed. Fresh **E02-modal-driver** R16 reaches exact $t = 10.0$, its production-window analyzer passes, R02 reaches $t = 2.5$, and the R17 timing lineage reaches $t = 0.10$. The **F-076** audit retains E02 only as pipeline and cost evidence because its forcing semantics are not exact. Thus the present manuscript establishes the method, target contract, and acceptance procedure, but does not yet claim reproduction of the MKS24 turbulent conclusions.

Completion first requires resolution of the passive-model fence, a qualified corrected executable, and then all sixteen mapped cases to reach $t_f \geq 10L_\perp/v_A$, pass strict safety and provenance checks, and be evaluated against each admitted comparison gate. Results of the later extension campaign will be reported only after that baseline is complete and a separate execution plan is approved.

18 Data and Reproducibility

The frozen Stage I matrix is stored in `inputs/cgl_lf_paper/mks24_stage_i_manifest.json`; preparation, submission accounting, and terminal-time acceptance inspection are performed by `scripts/frontier/cgl_lf_stage_i.py`. The archived E02-modal-driver production executable was built from revision `462b9dbd53e085dea46c2478b567781576d7d03e`, with archived executable SHA-256 digest `df87684e9d2b7af33b36c2757779d15de87b84ef051ca9c4489f2efa358f5c48`. For each segment, the production record retains the submitted input, executable and input provenance, Slurm allocation record, histories, snapshots, restarts, and an inspection record that tests target time and strict LF safety counters before acceptance. Production products are retained beneath the project `runs/mks24-stage-i` archive and will be linked to quantitative panel products only after their corresponding gates pass. Following the measured shared-I/O delay in job `4674731`, commit `1f52784d291718fff3bf1dc984a682ada52780ab` stores production snapshots and restarts as rank-local product sets; inspection retains every rank-file digest and analysis reconstructs the combined snapshot fields. Commits `476f9dbd`, `9c5c1b40`, and `bad8ba05` make analysis bundles follow inspected restart ancestry, permit the expected sampled output boundary without requiring duplicated history rows, and reject larger gaps inconsistent with the recorded history cadence. Commit `449e297b` records absent comparison cases explicitly during scoped single-case analysis and selects snapshot windows from lightweight binary-header reads before reconstructing retained fields.

19 Build and Runtime Status

The imported method figures come from the historical May 24, 2026 `full` bundle; the nonlinear figures come from the May 25, 2026 `paper-convergence` bundle. They have not been rerun for WO1. The current acceptance results are listed separately above; documentation compilation is a separate check from numerical validation. In the archived E02-modal-driver epoch, R16 reached exact $t = 10.0$, and its production-window analyzer passed. R02 reached exact $t = 2.5$ at the tracked F-075 checkpoint. The high-resolution R17 timing lineage reaches exact $t = 0.10$. F-076 preserves these as pipeline and cost evidence only. A corrected forcing-policy executable must be qualified before the mapped matrix starts again from $t = 0$, one inspected segment at a time, with R17 completed last. Production/statistical and paper-reference comparison gates remain open until the required corrected-epoch output bundles are analyzed against admitted references.

References

- [1] J. Squire et al., *Pressure anisotropy and viscous heating in weakly collisional plasma turbulence*, arXiv:2303.00468, 2023.
- [2] S. Majeski, M. W. Kunz, and J. Squire, *Self-organization in collisionless, high- β turbulence*, arXiv:2405.02418, 2024.